One less chip. Three signals.

Pico W + modified Deek-Robot L298P shield | Bridge A only

Remove the CD4077; omit the external level shifter. Keep the shield logic at 5 V and the planned motor supply at 15 V. Bridge A feeds both tracks in parallel; bridge B stays disabled.

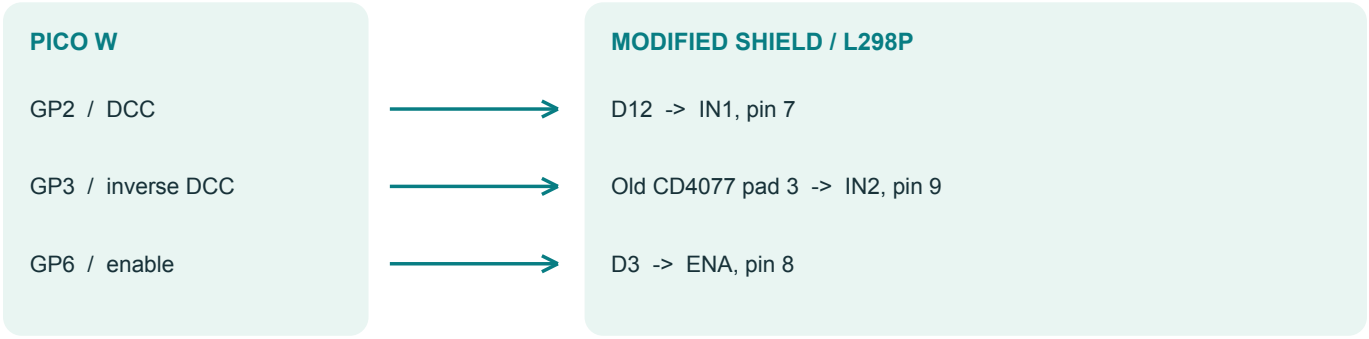
Firmware gate: release 0.5.3 produces GP2 only. This modified interface needs new firmware and bench verification before the tracks can be powered. This guide does not deploy firmware.

Why this works

The L298P accepts a HIGH from **2.3 V**, so Pico 3.3 V signals can drive it directly. Its logic supply still requires at least **4.5 V**. The CD4077 is the mismatch: at a 5 V supply it needs **3.5 V** for a guaranteed HIGH.

With braking disabled, the CD4077 inverts direction for the second bridge input. The Pico will instead generate **GP2/GP3 = 1/0, then 0/1** together in one PIO state machine.

Proposed signal path



Connection	Pico physical pin	Keep in the assembly
GP2 / GP3 / GP6	4 / 5 / 9	D3 / ENA: 10 kohm to GND
Common GND	3	D11 / ENB held low; B+ and B- unused

Package matters: L298P numbers above are for the surface-mount **PowerSO20**, not the upright L298N/Multiwatt. Confirm clone continuity before soldering.

Current sensing stays on A0 through the existing 10k/10k divider and 1 nF filter to GP26. Calibration and cutoff remain future work. Neither track rail is logic ground.

Document scope: this is the modified-board proposal. The existing Revision C KiCad drawing and BOM still describe the buffered alternative. Do not combine their signal wiring with this guide.

Modify, check, then power.

CD4077 removal and wiring | All connections must match the actual clone

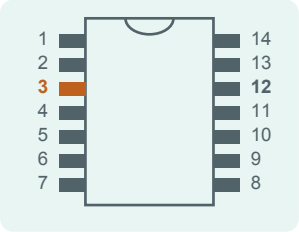
1. Verify the board with all power disconnected

Unplug USB, PD power and Pico/shield wiring. Photograph the board; identify the **CD4077** and its pin-1 marker. Keep the LM358 current-sense amplifier in place. Check these three continuity paths:

D12 -> L298P pin 7 | **CD4077 pin 3 -> L298P pin 9**
D3 -> L298P pin 8. If any path differs, trace the clone before proceeding.

2. Remove the CD4077 and connect the freed pad

Remove the entire CD4077 with suitable SMD rework tools. Clean and inspect the pads for bridges or lifted copper. Connect **GP3 to the PCB pad formerly occupied by CD4077 pin 3**, then add strain relief. Never drive this pad while the CD4077 output is still attached.



Top view, notch at the top.
Pad 3 is the third pad down on the left (orange). Use the actual pin-1 indicator and continuity, not a product photo's orientation.

Solder to the **board pad**; the chip has been removed.

Wire GP2 to D12, GP6 to D3 and common GND. Fit/check **10 kohm from D3 / ENA to GND** at the shield. Keep D11 / ENB low; B terminals unused. Retain the D8, D9 and D13 ground ties. Omit only the external AHCT buffer and its buffer-only wiring.

3. Verify before a locomotive goes on the rails

Power off: check the new wire reaches L298P IN2, with no accidental shorts to adjacent pads, 5 V or GND. Confirm the ENA pulldown and disabled ENB.

Logic power only; motor VIN disconnected: after the firmware update, verify complementary GP2/GP3 at the Pico and L298P inputs, with matching 58/100 us half-periods. ENA must stay low during startup, reset and OTA.

Power stage: after enable and current cutoff checks pass, use a current-limited supply and dummy load before a locomotive. Do not connect the generic logic analyzer directly to powered track outputs.

Firmware work still required

Generate both outputs together; enable only after valid DCC starts; disable ENA before stopping PIO, during OTA and on faults; implement/calibrate current cutoff and startup speed-zero packets. **These controls are not implemented in release 0.5.3.**

Primary references

Arduino Motor Shield Rev3 schematic - Reference routing; clone must be checked.

ST L298 datasheet, pages 4-5 - PowerSO20 pinout and supply/input ratings.

TI CD4077B datasheet, pages 2-4 - Pinout and 5 V input threshold.

Use at your own discretion. The magic smoke is not a commissioning indicator.